Taylor W. Killian

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Research Interests

Techniques for efficient and effective *sequential decision making* in the presence of uncertainty: particularly toward improved *reliability* and *generalization*, motivated by real world challenges.

Subject Areas

- Reinforcement Learning
  - Offline RL
  - Risk-sensitive RL
  - Model-based RL
- LLMs, Reasoning & Agents
- Representation Learning
- Causal Inference
- Approximate Bayesian Inference
- Health and Wellness

Education

Ph.D., Computer Science **April 2024**
University of Toronto, Toronto, ON, Canada
- GPA: 4.0

Advisor: Marzyeh Ghassemi
Thesis: Clinically Motivated Sequential Decision Making under Uncertainty in Offline Settings

M.Eng, Computational Science and Engineering **May 2017**
Harvard University, Cambridge, MA
- GPA: 3.92

Advisor: Finale Doshi-Velez
Thesis: Robust and Efficient Transfer Learning by Accounting for and Modeling Parameter Variation

BS, Mathematics **April 2013**
Brigham Young University, Provo, Utah
- GPA: 3.83

Skills and Languages

- Python
- Tensorflow, Pytorch, Jax
- LaTeX, MATLAB
- Java, CUDA, C++
- Fluent in Swedish

Awards

- TMLR Expert Reviewer
- ICML, NeurIPS, ICLR: Top Reviewer Award, 2019 - 22
- AAAI, Student Abstract Best Presentation, 2017
- MIT LL Lincoln Scholar, 2015-2017
- NDSEG Fellowship Award, 2013 (Declined)
- SMART Fellowship Finalist, 2011
- BYU ORCA Grant Recipient, 2010

Experience

Assistant Professor, Department of Computer Science **August 2026 - Present**
Brigham Young University
- My research group is focused on developing adaptive decision making algorithms for pragmatic Machine Learning use cases
- Developing Machine Learning curriculum, especially for Reinforcement Learning
- Responsible for BYU CS's Graduate Research Seminar, supporting our Graduate Committee
- Faculty advisor for Student-led AI Association, focused on expanding campus wide AI Literacy

Member of Technical Staff, AI Research **March 2026 - Present**
Lila Sciences

- Providing on-going consulting for post-training efforts of Scientific Reasoning Models
- Developing advantageous research directions in distillation, RL, and temporal abstraction.

Senior Research Scientist, RL and Post-training Lead **February 2025 - February 2026**
MBZUAI, Institute of Foundation Models

- Founding member of North American extension of MBZUAI research entity.
- Developed open-source techniques to facilitate general LLM reasoning capabilities.
  - Co-led development of K2-Think, a highly capable 32B parameter reasoning model

Postdoctoral Research Scientist **December 2023 - December 2024**
Apple Inc., Special Projects Group – Physical Intelligence

- Established goal-conditioned, contextual RL policies in self-play for multi-agent autonomy.
- Developed solutions for decision-making in partially observed, interactive settings.
- Explored personalization of large-language model agents in long-context settings.

Graduate Research Assistant **August 2019 - April 2024**
Department of Computer Science, University of Toronto & Vector Institute
Institute of Medical Engineering and Science, Massachusetts Institute of Technology

- Developed robust representations of patient health, incorporating measures of uncertainty
- Introduced method to identify and avoid detrimental decisions in high-risk environments
- Pursued research to enable robust knowledge transfer between healthcare institutions by combining causal inference and reinforcement learning

Research Intern **May 2023 - September 2023**
Apple Inc., AI/ML – Health AI *supervised by Nick Foti*

- Developed framework to efficiently adapt global recommendation policies to local contexts.
- Instituted agenda to integrate novel and contemporary RL approaches for Apple Health AI.

Research Intern **June 2022 - September 2022**
Microsoft Research, Health Futures – BioML *supervised by Ava Amini*

- Extended distribution-free uncertainty quantification to continuous-time sequential settings
- Developed mechanism to estimate prediction confidence among irregularly sampled time-series.

Research Intern **June 2021 - December 2021**
Apple Inc., Health AI *supervised by Leon Gatys and Joern-Henrik Jacobsen*

- Explored self-supervised learning for representation of user health via physiological signals
- Served as in-house technical expert on sequential decision making approaches

Student Researcher / Research Intern **May 2020 - December 2020**
Google Research, Brain team *supervised by Marlos Machado and Marc Bellemare*

- Studied the utility of embedding uncertainty measures in an RL agent's state representation

Teaching Assistant **August 2019 - May 2020**
Department of Computer Science, University of Toronto

- Part of teaching staff for:
  - CSC311 Introduction to Machine Learning
  - CSC2541 Machine Learning for Healthcare
- Developed and administered assignments and exams
- Worked with and instructed students, answering questions about course material
- Coordinated projects with clinical collaborators, organized and advised student groups

Associate Technical Staff **June 2017 - July 2019**
Air, Missile and Maritime Defense Technology, MIT Lincoln Laboratory

- Led effort to identify and develop areas for Laboratory investment in Artificial Intelligence
- Developed ML algorithms for efficient and accurate performance in low-data regimes
- Fused multiple information sources to reduce false-alarms in aviation passenger screening

Assistant Technical Staff **May 2013 - May 2017**
Air, Missile and Maritime Defense Technology, MIT Lincoln Laboratory

- Evaluated the impact of technological and operational improvements to U.S. missile defenses
- Developed and performed data-driven analyses to identify U.S. DoD capability improvements
- Produced briefing materials to present to key U.S. DoD decision makers

Undergraduate Research Assistant **June 2010 - May 2013**
Department of Mechanical Engineering, Brigham Young University

- Published research on fluid activated passive dampening as primary author
- Trained in methods of applied mathematics partnered with computation in MATLAB
- Furnished analytical and mathematical support to experimental techniques

Office of Naval Research NREIP Intern **Summer 2011**
Naval Surface Warfare Center, Dahlgren, Virginia

- Designed and carried out experiment to study optimal nose shape for submerging projectiles
- Presented results to division commanding officers and staff.
- Obtained U.S. Department of Defense security clearance.

Language Instructor (Swedish) **December 2009 - June 2010**
Missionary Training Center, Church of Jesus Christ of Latter-day Saints

- Prepared lessons to satisfy language and theological curriculum and individual student needs.
- Counseled with students to overcome individual concerns and problems.
- Instructed and evaluated language fluency and understanding of concepts.

Publications

- Zhao, K., Yu, T., Li, S., Yang, Y., Hughes, A., Zhang, W., **Killian, T.** "**Recovering Reliable Supervision via Critique-Guided On-Policy Distillation**". *in Submission.*
- Yang, Y., Li, S., Yu, T., Zhao, K., **Killian, T.**, Zhang, W. "**REVO: Rollout-Efficient Off-Policy Distillation via Variance-Guided Reuse**". *in Submission.*
- Yu, T., Zhao, K., Li, S., Yang, Y., Jenkins, P., Zhang, W., **Killian, T.** "**Rethinking Training-Inference Mismatch in LLM Reinforcement Learning: Where It Arises and How to Correct It**". *in Submission.*
- Yang, Y., Yu, T., Li, S., Zhao, K., Zhang, X., Bansal, C., Yao, H., **Killian, T.**, Zhang, W. "**When EOS Tokens Disagree: Understanding Length Inflation in On-Policy Distillation**". *in Submission.*
- Cui, C., **Killian, T.**, Ammanabrolu, P. "**Behavior Cue Reasoning: Monitorable Reasoning Improves Efficiency and Safety through Oversight**". *in Submission.*
- Deng, M., Hou, J., Sa Neves, L., Pimpalkhute, V., **Killian, T.**, Liu, Z., Xing, E., "**Efficient Agentic Reasoning through Self-Regulated Simulative Planning'**". *in Submission*
- Gao, C., Li, H., **Killian, T.**, She, J., Wang, R., Ma, L., Cheng, Z., Hao, S., Xu, Z., "Concise Reasoning in the Lens of Lagrangian Optimization". *to Appear: Advances in Neural Information Processing Systems (NeurIPS).* December 2026.
- Yu., T., Zhao, K., Chen, C., Hughes, A., **Killian, T.**, Ma, F., Zhang, W., Jenkins, P. "**LARK: Learnability-Grounded Trajectory Selection for Efficient Reasoning Distillation**". *to Appear: Advances in Neural Information Processing Systems (NeurIPS).* December 2026.
- Kong, L., Liu, X., Chen, G., Ma, M., Song, X., Sun, Y., Yurochkin, M., **Killian, T.**, Salakhutdinov, R., Zhang, K., Xing, E., Liu, Z. "**From Reasoning Traces to Reusable Modules: Understanding Compositional Generalization in Language Model Reasoning**". *International Conference on Machine Learning (ICML).* July 2026.
- Cheng, Z., Xie, Y., Qu, Y., Setlur, A., Hao, S., Pimpalkhute, V., Liang, T., Yao, F., Liu, Z., Xing, E., Smith, V., Salakhutdinov, R., Hu, Z., **Killian, T.**, Kumar, A., "**IsoCompute Playbook: Optimally Scaling Sampling Compute for LLM RL**". *International Conference on Machine Learning (ICML).* July 2026.
- Landers, M., **Killian, T.**, Hartvigsen, T, Doryab, A., "**Improving and Accelerating Offline RL in Large Discrete Action Spaces with Structured Policy Initialization**". *International Conference on Learning Representations (ICLR).* May 2026.
- Landers, M., **Killian, T.**, Hartvigsen, T, Doryab, A., "**SAINT: Attention-Based Modeling of Sub-Action Dependencies in Multi-Action Policies'**". *arxiv 2505.12109*
- Landers, M., **Killian, T.**, Barnes, H., Hartvigsen, T, Doryab, A., "**BraVE: Offline Reinforcement Learning for Discrete Combinatorial Action Spaces**". *Advances in Neural Information Processing Systems (NeurIPS).* December 2025.
- Liu, Z., Tang, L., Jin, L., Li, H., Ranjan, N., Fan, D., Rohatgi, S., Fan, R., Pangarkar, O., Wang, H., Cheng, Z., Sun, S., Han, S., Tan, B., Gosal, G., Han, X., Pimpalkhute, V., Hao, S., Hee, M.S., Hestness, J., Jia, H., Ma, L., Singh, A., Soboleva, D., Vassileva, N., Wang, R., Wu, Y., Sun, Y., **Killian, T.**, Moreno, A., Maggs, J., Ren, H., He, G., Wang, H., Ma, X., Wang, Y., Yurochkin, M., Xing, E. "**K2-v2: A 360-open, reasoning-enhanced LLM**". *Technical Report.* December 2025. *arxiv 2512.06201.*
- Cheng, Z., Hao, S., Liu, T., Zhou, F., Xie, Y., Yao, F., Bian, Y., Zhuang, Y., Dey, N., Zha, Y., Gu, Y., Zhou, K., Wang, Y., Li, Y., Fan, R., She, J., Gao, C., Saparov, A., Li, H., **Killian, T.**, Yurochkin, M., Liu, Z., Xing, E., Hu, Z., "**Revisiting Reinforcement Learning for LLM Reasoning from A Cross-Domain Perspective**", *Advances in Neural Information Processing Systems (NeurIPS).* December 2025.

- Cheng, Z., Fan, R., Hao, S., **Killian, T.**, Li, H., Sun, S., . . . , He, G., Hu, Z., Liu, Z., Xing, E., "**K2-Think: A Parameter-Efficient Reasoning System**". *Technical Report.* September 2025. *arxiv 2509.07604*
- Cusmano-Towner, M., Hafner, D., Hertzberg, A., Huval, B., Petrenko, A., Vinitsky, E., Wijmans, E., **Killian, T.**, Bowers, S., Sener, O., Krähenbühl, P., Koltun, V., "**Robust Autonomy Emerges from Self-Play**", *International Conference on Machine Learning (ICML).* July 2025.
- **Killian, T.**, Zhang, H., Hartvigsen, T., Amini, A., "**Continuous Time Evidential Distributions for Irregular Time Series**", *arXiv 2307.13503*
- Casper, S., **Killian, T.**, Hadfield-Menell, D., Kreiman, G., "**White-Box Adversarial Policies in Deep Reinforcement Learning**". *arXiv 2209.02167*
- **Killian, T.**, Parbhoo, S., Ghassemi, M., "**Risk-Sensitive Dead-end Identification in Safety-Critical Offline Reinforcement Learning**", in *Transactions on Machine Learning Research.* January 2023. https://openreview.net/forum?id=oKlEOT83gI.
- **Killian, T.**, Ghassemi, M., Joshi, S., "**Counterfactually Guided Off-policy Transfer in Clinical Settings**", *Conference on Health, Inference and Learning (CHIL).* April 2022.
- Fatemi, M., **Killian, T.**, Subramanian, J., Ghassemi, M., "**Medical Dead-ends and Learning to Identify High-Risk States and Treatments**", *Advances in Neural Information Processing Systems.* December 2021
- **Killian, T.**, Zhang, H., Subramanian, J., Fatemi, M., Ghassemi, M., "**An Empirical Study of Representation Learning for Reinforcement Learning in Healthcare**" *Machine Learning for Health Workshop, NeurIPS,* December 2020
- D'Costa, A., Denkovski, S., Malyska, M., Moon, S.Y., Rufino, B., Yang, Z., **Killian, T.**, Ghassemi, M., "**Multiple Sclerosis Severity Classification From Clinical Text**", *The 3rd Clinical Natural Language Processing Workshop*, EMNLP 2020.
- **Killian, T.**, Subramanian, J., Fatemi, M., Ghassemi, M., "**Learning Representations for Prediction of Next Patient State**", *1st Annual ACM Conference on Health, Inference and Learning*, April 2020. **Workshop Spotlight**
- Silva, A., **Killian, T.**, Jimenez, I., Son, S.-H., Gombolay, M. "**Optimization Methods for Interpretable Differentiable Decision Trees Applied to Reinforcement Learning**", *The 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*, August 2020.
- **Killian, T.**, Goodwin, J., Brown, O. & Son, S-H., "**Kernelized Capsule Networks**", *1st Workshop on Understanding and Improving Generalization in Deep Learning, ICML*, July 2019.
- Yao, J., **Killian, T.**, Konidaris, G. & Doshi-Velez, F., "**Direct Policy Transfer via Hidden Parameter Markov Decision Processes**", *The 2nd Lifelong Learning: A Reinforcement Learning Approach Workshop, ICML*, July 2018. **Selected for Oral presentation.**
- Jones, A., **Killian, T.**, Hurley, M., & Allen, R., "**Artificial Intelligence and Machine Learning for Decision Support: Recommendations for Investment**", **Technical Report**, MIT Lincoln Laboratory, June 2018. *Paper not available for public release*
- **Killian, T.**, Daulton, S., Konidaris, G. & Doshi-Velez, F. , "**Robust and Efficient Transfer Learning in Hidden Parameter Markov Decision Processes**", *Advances in Neural Information Processing Systems.* December 2017 **Selected for an Oral presentation**
- **Killian, T.**, Konidaris, G. & Doshi-Velez, F., "**Robust and Efficient Transfer Learning in Hidden Parameter Markov Decision Processes**". In *AAAI* (pp. 4949-4950). February 2017.
- **Killian, T.**, Klaus, R. & Truscott, T.T., "**Rebound and jet formation of a fluid-filled sphere**", *Physics of Fluids* **24**, 122106 (2012), DOI:10.1063/1.4771985.

Invited Talks

- *28 October 2026 – Invited Talk*, **Cohere for AI**
  "Decision-Aware Measurement Selection"
- *6 October 2026 – Invited Talk*, **BYU AI Association**
  "A Professor's Journey in Reinforcement Learning"
- *11 September 2026 – Invited Talk*, **BYU CS Weekly Graduate Research Seminar**
  "How to Make a BYU Professor"
- *11 August 2026 – Invited Talk*, **Real World RL Workshop; Hanover, Germany**
  "Creative Usage of Rewards for Real World RL"
- *21 January 2026 – Invited Talk*, **Lila Sciences**
  "Fully Open LLM Reasoning at the Frontier"
- *8 May 2025 – Invited Talk*, **Brigham Young University, Computer Science Dept.**
  "Pragmatic Reinforcement Learning for Real-World Use"
- *4 February 2025 – Invited Talk*, **Brigham Young University, Math Dept.**
  "Pragmatic Reinforcement Learning in Safety-Critical Domains"
- *10 January 2025 – Invited Talk*, **Genentech, gCS RED**
  "Clinically Motivated, Pragmatic Reinforcement Learning"
- *9 January 2025 – Invited Talk*, **Harmonic**
  "Pragmatic, Uncertainty-guided Reinforcement Learning"
- *8 January 2025 – Invited Talk*, **Oracle Labs**
  "Sequential Decision Making under Uncertainty in Safety-Critical Scenarios"
- *3 December 2024 – Invited Talk*, **Google Research, Health AI**
  "Practical Reinforcement Learning in Healthcare"
- *4 November 2024 – Invited Talk*, **RBC Borealis, HELM Team**
  "Pragmatic, Uncertainty Guided Reinforcement Learning"
- *24 October 2024 – Invited Talk*, **Meta, Applied Reinforcement Learning Team**
  "Practical Reinforcement Learning in Safety-Critical Domains"
- *19 April 2024 – Invited Talk*, **Stanford University – Brunskill Group**
  "Pragmatic, Uncertainty Guided Reinforcement Learning"
- *22 September 2023 – Invited Talk*, **Brown University – bigAI**
  "Pragmatic, Uncertainty Guided Reinforcement Learning"
- *13 September 2023 – Invited Talk*, **NYU Center of Data Science**
  "A Pragmatic View of Offline Reinforcement Learning in Safety-Critical Scenarios"
- *22 August 2023 – Invited Talk*, **Distributional RL Book Launch, MILA**
  "Distributional RL for Safety-Critical Offline Settings"
- *20 April 2023 – Invited Talk*, **Apple, Health AI**
  "A Pragmatic View of Reinforcement Learning in Healthcare"
- *2 December 2022 – Invited Talk*, **NeurIPS 2022 Offline RL Workshop**
  "Identification of Dead-ends in Safety-Critical Offline RL"
- *17 November 2022 – Guest Lecture*, **MIT 6.7950 – RL: Foundations And Methods**
  "Context Matters: Leveraging Latent Information to Solve Families of MDPs"
- *10 July 2020 – Invited Talk*, **Vector Institute Friday Seminar** (virtual)
  "Counterfactually Guided Policy Transfer in Clinical Settings"
- *20 February 2020 – Guest Lecture*, **UofToronto CSC 2541 – ML for Healthcare**
  "Reinforcement Learning for Healthcare"

- *18 October 2017 – Guest Lecture*, **Harvard CS 282R – RL for Healthcare**
  "Unwinding the DQN: Tips and Tricks"
- *12 June 2017 – Invited Talk*, **SMG: Boston, Site Visit**
  "Experiences with Computational Science at Harvard"

Teaching

- CS 270 – Introduction to Machine Learning — Brigham Young University, Winter 2027
- CS 401R – Introduction to Reinforcement Learning — Brigham Young University, Fall 2026
- CSC 2541 – ML for Healthcare (Lead TA) — University of Toronto, Winter 2020
- CSC 311 – Introduction to Machine Learning (TA) — University of Toronto, Fall 2019
- CS 282R – RL for Healthcare (Guest Lecture) — Harvard University, Fall 2017

Thesis/Dissertation Committees

- Serving on several thesis/dissertation committees; all are currently in progress.

Volunteer

Program Committee/Reviewer

- **ICML**: 2019-Present
- **NeurIPS**: 2018-Present
- **CHIL**: 2020-Present
- **ICLR**: 2020-Present
- **TMLR**: 2022-Present
- **JMLR**: 2021-Present
- **NeurIPS ML4H Workshop**: 2017-Present
- **NeurIPS Offline RL Workshop**: 2020-Present
- **NeurIPS Deep RL Workshop**: 2020-Present
- **RLC**: 2023-Present

Graduate Committee **Aug 2026 - Present**
BYU Computer Science Department

- Organizing weekly Graduate Research Seminar
- Contributing to decisions about adapting Graduate program curriculum and admissions

Program Committee – 2026 RLC Workshop Chair **2025-2026**

- Solicited, evaluated and selected proposals for workshops to be organized at RLC 2026
- Led and facilitated more rigorous review and evaluation of workshop proposals

Organization Committee – 2024 RLC Workshops **2024-2025**
"I Can't Beleive It's Not Better!": Failure Modes of Sequential Decision-Making in Practice
Finding the Frame: An RLC Workshop for Examining Conceptual Frameworks in RL

- Helped develop the successful proposal of two workshops for the inaugural RL Conference.
- Led ideation and writing of proposals, each focusing on unique points of failure in current research practices in RL
  - First, looking into ideas/concepts that do not scale or otherwise transfer from simulated, toy domains to real-world practice.
  - Second, identifying problematic dogmas in conceptual frameworks for RL and encouraging alternative formulations to unlock new algorithmic capabilities or intended use cases.

Virtualization Chair **2021-2022**
Machine Learning for Healthcare Symposium

- Helped lead the development of a standalone symposium, co-located with NeurIPS, in response to growth in interest among the research community

- Led the planning and execution of both a fully virtual event (2021) and hybrid hosting (2022)

Technical Recruiter, Campus Recruiting **December 2014 - July 2019**
Human Resources, MIT Lincoln Laboratory

- Organized campus information events to introduce Laboratory mission and aims
- Served as mentor and advocate for candidates seeking employment

Committee Member; PED Seminar Series **June 2015 - December 2017**
MIT Lincoln Laboratory

- Assisted in the organization and planning of seminar series focused on leveraging modern computation techniques to extract actionable insight
- Hosted leading researchers in Machine Learning and Artificial Intelligence

President, Student Advisory Committee **January 2011-December 2011**
Department of Mathematics, Brigham Young University

- Led effort to improve curriculum and increase collaboration between students and faculty.
- Planned and carried out activities to promote mathematics and educate BYU community.

Missionary, Sweden Stockholm Mission **March 2007 - March 2009**
The Church of Jesus Christ of Latter-day Saints

- Managed and oversaw the work and safety of 12 other missionaries in remote areas of Sweden.
- Trained 6 newly arrived missionaries in language, culture, and proselyting skills.
- Served full-time as a church representative identifying and meeting the needs of the community.